

Differential Equations and Slope Fields



Verifying solutions to differential equations

- 1 Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C .
 - 2 Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.
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Solving separable differential equations

- 3 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
 - 4 Solve $\frac{dy}{dx} = 2xy$ given $y(0) = 1$.
 - 5 Solve $\frac{dy}{dx} = \frac{\cos x}{y^2}$ given $y(0) = 2$.
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Slope fields

- 6 For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, as would be drawn on a slope field.
 - 7 Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.
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Exponential growth and decay models

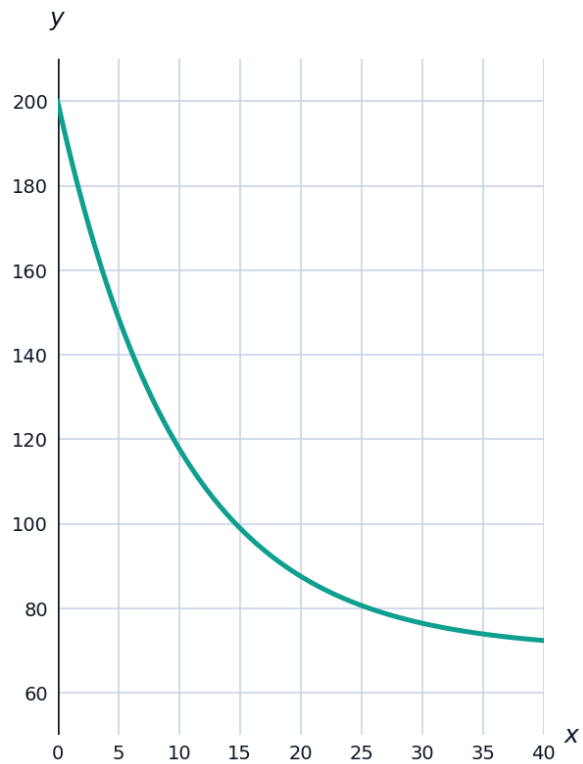
- 8 A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.
 - 9 A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).
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Matching slope fields to differential equations

- 10 A slope field shows horizontal segments (slope 0) along the entire x -axis and along the entire y -axis, with slopes elsewhere resembling the product xy . Identify the differential equation this slope field represents.
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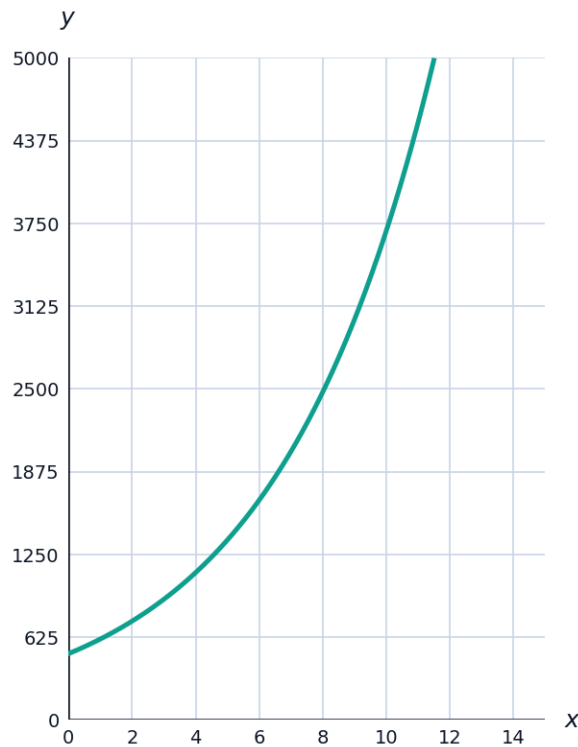
Newton's Law of Cooling

- 11** An object cools according to $\frac{dT}{dt} = -k(T - 70)$, where 70°F is room temperature. If $T(0) = 200^\circ\text{F}$ and $k = 0.1$, find $T(t)$.



Visualizing an exponential growth solution

- 12 The graph shows the bacteria population $P(t) = 500e^{0.2t}$ from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using $\ln 2$.



A logistic-style differential equation

- 13 A population satisfies $\frac{dP}{dt} = 0.1P\left(1 - \frac{P}{200}\right)$. Find the equilibrium solutions (where $\frac{dP}{dt} = 0$), and explain which one is stable for a population starting near $P = 10$.

Additional separable equations

- 14 Solve $\frac{dy}{dx} = \frac{y}{x}$ given $y(1) = 6$.
- 15 Solve $\frac{dy}{dx} = \frac{1}{y}$ given $y(0) = 4$.

A mixing problem modeled by a differential equation

- 16 A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If $y(t)$ is the amount of salt (kg) at time t , the differential equation is $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$. Find the equilibrium value of y , and explain what it represents physically.